Tailoring and tuning of the broadband spectrum of a step-chirped grating based frequency doubler using tightly-focused Gaussian beams

Ameneh Bostani,[1,*] Meenu Ahlawat,[1] Amirhossein Tehranchi,[2,3] Roberto Morandotti,[3] and Raman Kashyap[1,2]

[1]*Dept. of Engineering Physics, Ecole Polytechnique de Montreal, Montreal, QC, H3C 3A7, Canada*
[2]*Dept. of Electrical Engineering, Ecole Polytechnique de Montreal, Montreal, QC, H3C 3A7, Canada*
[3]*INRS-EMT, 1650 Boulevard Lionel Boulet, Varennes, QC, J3X 1S2, Canada*
[*]*bostani.ameneh@polymtl.ca*

Abstract: We demonstrate theoretically and experimentally, that the non-uniform spectra of second harmonic generation (SHG) from an unapodized step-chirped periodically poled nonlinear optical grating can be apodized utilizing tightly-focused Gaussian beams to suppress the ripple in its wideband response. In our example, by increasing focusing, a ripple-free response is progressively achieved over a 6-dB bandwidth of >5nm, with a beam waist of 20 μm. With this tight focusing arrangement, a continuous tuning of 11-nm is also demonstrated by simply changing the focal point by 5.8 mm within the step-chirped grating based APPLN.

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1. Introduction

Ultrafast optical signal processing based on cascaded second-order nonlinearities requires a broadband frequency doubler to perform waveband wavelength conversion [1–6] A good candidate for this purpose is quasi-phase matched second-harmonic generation (SHG) in poled crystals [7,8]. As a uniform, periodically poled material has a narrow bandwidth, chirped gratings [9] and step-chirped gratings [10,11] in the form of aperiodically poled material on the other hand, have been proposed to realize a broad bandwidth for SHG. However the SH response of these devices suffer from a highly non-uniform spectral response. Therefore, apodization techniques are applied during fabrication to minimize the ripples in the SH spectrum [10–12]. These methods are based on altering the effective nonlinearity to zero progressively by decreasing the duty cycle, i.e., the length ratio of positively and negatively poled regions, at the edges of chirped device [11–13]. Applying this kind of apodization with the fabrication of extremely fine-poled regions requires very precise and advanced equipment. To reduce the ripples, we suggest a new scheme for introducing some apodization, by launching tightly focused light into a step-chirped (SC) aperiodically poled material such as lithium niobate and controlling the wavelength range of the spectrum by moving its focal point inside the device. This technique also can be used in an apodized structure with a duty ratio variation to arrive at a smoother response and possibly a larger bandwidth as well.

The role of a focused light beam for frequency mixing for SHG in a nonlinear crystal [14–16] and in periodic structures, in particular PPLN, has been investigated theoretically and experimentally to determine the optimum focusing conditions [17–19]. Using a tighter focus in a periodic structure induces a Gouy phase shift [19], which reduces the maximum efficiency in comparison with a plane wave. Also a new design has been suggested to correct this phase in the periodic structure [17]. The effect of adding chirp and Gouy phase shift compensator to a periodic structure has been studied using local phase mismatch in chirped and uniform structures for a Gaussian beam in order to find an optimum chirp rate to maximize efficiency [20]. The dependence of SHG on the focusing position has also been investigated in non periodically poled nonlinear crystals indicating that the maximum power is not generally yielded in the central focal point [21]. However, the effect of focusing and the focal position in the crystal have not been investigated to the best of our knowledge, in a chirped structure for frequency doubling. Of course, this applies to the generalized case of frequency mixing as well.

In this paper, in section 2, we discuss the theory of Gaussian beams for engineering the SHG response, which is used in our simulations. In section 3, the special design and fabrication of a wideband frequency doubler in the form of an unapodized step-chirped (SC) aperiodically poled lithium niobate (APPLN) are described. In section 4 we demonstrate the possibility of control over the SH efficiency of an unapodized SC-APPLN for different fundamental harmonic (FH) wavelengths obtained by focusing the FH pump to different beam waists. This allows us to engineer the conversion efficiency profile. In section 5 we examine the influence of shifting the focal point within the SC-APPLN to tune the spectrum

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of the SHG response. The experimental results are also compared with the theoretical simulations.

2. Theory

The coupled wave equations for SHG in chirped gratings can be written as [9]

$$\frac{dA_1}{dz} = \frac{2j\omega_1^2 d(z)}{n_1 c} A_2 A_1^* e^{-j\Delta kz}$$
$$\frac{dA_2}{dz} = \frac{2j\omega_2^2 d(z)}{n_2 c} A_1^2 e^{j\Delta kz} \tag{1}$$

where A_1 and A_2 are the fields of FH and second harmonic (SH) which are perpendicular to propagation direction (z), $\Delta k = k_{2\omega} - 2k_{\omega}$ is the phase mismatch between the SH and FH. c is the speed of light, $\omega_{1,2}$, $n_{1,2}$ and $k_{1,2}$ are the angular frequencies, refractive indexes and wave numbers of FH and SH field. $d(z) = f_{ap}(z) d_{chirped}$ is the nonlinearity of structure composed of two terms of apodization (f_{ap}) and chirping ($d_{chirped}$). An apodization function is used to bring the effective second order nonlinear coefficient to zero at the edges of the grating, thereby diminishing the side lobes of a sinc-type response function in unchirped gratings as well as the ripples in chirped gratings [10, 11]. Chirped structures have a monotonically variable period to cover all required periods for broadband SHG. The step-chirped grating consists sections, each of uniform gratings with different periods. The period in the i'th section changes as

$$\Lambda_i = \Lambda_1 + (i-1)\Delta, \tag{2}$$

where Λ_1 is the period of first section and Δ represents the period difference between two successive sections.

When a laser beam is focused in the crystal, the electric field contribution using a first order approximation of electric field component results in [22]

$$E_{1y}(z) = \frac{E_0}{1+i\zeta} \exp\left(-\frac{x^2+y^2}{\omega_0^2(1+i\zeta)}\right) e^{(ikz)}, \tag{3}$$

where $\zeta = 2(z-f)/b$ where f is the focal position and $b = 2\pi\omega_0^2 n_1/\lambda_1$ is the confocal parameter. ω_0, λ_1 and E_0 are beam waist and FH wavelength and initial electric field ($|E(0)|$), respectively. Considering the variation in the non-depleted case, the total SH field contribution at the observation point (r', z') outside the crystal can be written as

$$E_2(r', z') = \frac{ik_2 E_0}{4n_2} \exp\left(-\frac{2(x^2+y^2)}{\omega_0^2(1+i\zeta')}\right) \int \left[\frac{d(z)}{1+i\zeta(z)}\right] \exp(i\Delta kz) dz, \tag{4}$$

where x, y, z are considered to be the coordinates of the source [15]. The first term in brackets can be considered as the total nonlinearity.

3. Design and fabrication

In order to design a wideband frequency doubler based on chirped gratings in PPLN with an SH efficiency centered at the FH wavelength of 1550 nm and possessing a bandwidth of 30 nm at a temperature of 125 °C, the period of the device should change from 18.209 μm to 19.019 μm. The device covers SHG between 1535 nm and 1565 nm. However, achieving the resolution below 100 nm makes the mask fabrication for the photolithography process difficult and expensive. Therefore we designed an unapodized SC-APPLN considering this

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limitation and its schematic is shown in Fig. 1(a). The total grating length (L) is divided into 10 equal sections with a varying periodic pattern. Using Eq. (2) with Λ_1 = 18.2 μm and Δ = 0.1 μm, the last period is Λ_{10} = 19.1 μm. The device efficiency is then the power ratio of the output SH to input FH at any wavelength. The total length is engineered to be 20 mm, a compromise between the efficiency and the ripples. The uniform, 2-mm-long sections possess a 5-nm bandwidth which are short enough to produce a continuous 30-nm bandwidth with < ~6 dB peak-to-peak ripple.

Fabrication of the SC-APPLN is based on electric field poling at room temperature, a common technique used for the fabrication of QPM devices, especially for LN [23]. The + z face of an undoped, 0.5-mm-thick, z-cut, optical grade lithium niobate crystals were used for the fabrication. The LN wafers were then coated with a 1.5 μm layer of positive photo-resist S1813 and exposed to 314 nm UV radiations through a specially designed SC grating pattern on the mask. After lithographic printing, the sample was mounted on a holder contacting both surfaces with saturated lithium chloride (LiCl) solutions as the electrolyte. A dc-electric field (22 kV/mm), higher than the coercive field strength of LN in the shape of pulses with duration of 0.5 ms were applied to the liquid electrode pattern until domain inversion is achieved in all grating regions. The in situ monitoring of the poling process was made using a video camera through a crossed polarizer setup, observing the induced birefringence change between the poled and un-poled regions. The duty cycle and period of APPLN were examined under microscope after cleaning and etching in hydro⟩uoric acid (HF 49%). A microscope image of a section of the fabricated device is shown in Fig. 1(b).

In uniform gratings, domain broadening through fabrication errors reduces the efficiency. However, in an SC-APPLN, the broadening may change the shape of the spectrum and ripple depth if it is not exactly the same in each section. The response of the fabricated SC-APPLN can turn out to be better in ripple depth than designed SC-APPLN as the overall response of the fabricated one comes from the random phase accumulation of the SH electric field of each section. Our simulations indicate that if there is a random broadening in order of 20% of grating period, it influences the ripple depth less than 15%.

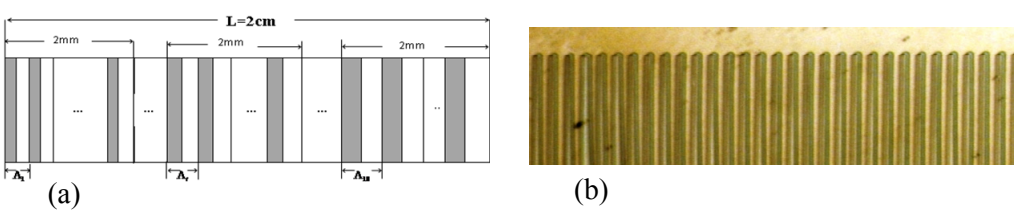

Fig. 1. (a) Schematic of the designed SC-APPLN. (b) Part of the fabricated sample viewed under microscope after etching.

4. SH Spectrum tailoring by focusing with different beam waists

In order to examine the effect of focusing on the wideband SH response of a step-chirped grating device, we simulated SHG with a Gaussian beam of different beam waists in the specially-designed SC-APPLN, and verified it experimentally.

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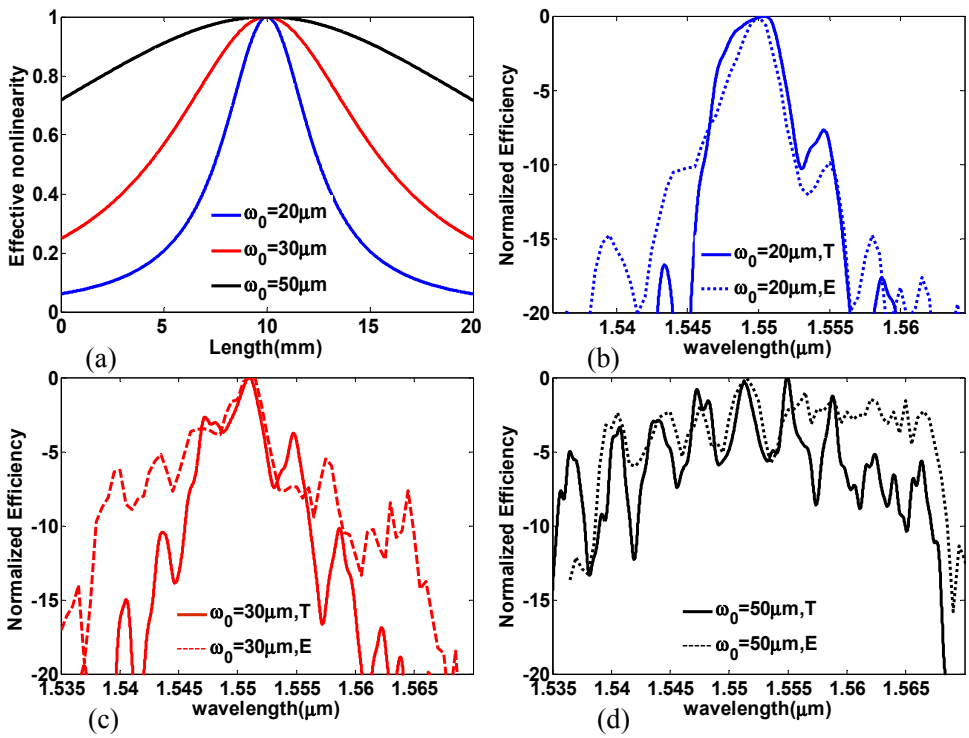

Fig. 2. (a) Normalized effective second-order nonlinearity vs. grating length as a function of focused beam waist and (b-d) Normalized measured and simulated SHG efficiency vs. FH wavelength, for a focused light with different beam waists. In the legend "T" and "E" represent theoretical and experimental result, respectively.

In our theoretical simulations, the output power is calculated by integrating a Gaussian beam over the cross section as it changes along the length of a crystal. In the step-chirped gratings $d(z) = d_{chirped}(z)$ with focusing, part of the electric field variation can be translated to a change of the effective second-order nonlinearity and thus considered as an apodization term in the poled crystals as $f_{ap}(z) = 1/(1+i\zeta)$. This function is shown in Fig. 2(a) for different beam waists (ω_0) of the focused light versus the length of crystal. In this plot, the focal position is considered to be in the middle of crystal; however, by changing this position the maximum in each graph changes to the point of the focus.

In the experimental setup, three different lenses (f = 5 cm, 7 cm, 12.5 cm) were used to achieve a focused beam waist of 20, 30 and 50 microns, respectively, in the center of the SC-APPLN. A JDSU tunable laser was used as the pump in the C band, amplified by a Pritel high-power EDFA with M^2 value of ~1.0 and passed through a polarization controller before being focused into the center of the SC-APPLN. The SC-APPLN sample was placed in a temperature-controlled oven at a fixed temperature of 125° C to avoid thermal damage. The output FH and SH waves were evaluated using an optical spectrum analyzer and a power meter. For an input pump power, P_p = 0.66 W, the experimentally observed peak power is ~0.21 mW which is in good agreement with the theoretical calculation, P_{SHG} = 0.22 mW, resulting in an efficiency of −34.78 dB. The normalized SHG efficiency versus FH wavelength is plotted in Figs. 2(b-d) and compared with the simulation of a perfect SC-APPLN structure. We found, somewhat surprisingly, that the efficiency profiles for various beam waists change dramatically due to the apodization resulting from focusing. We chose a bandwidth of 6-dB due to the large ripples in the response of the SC-APPLN, in order to show how significantly the ripples reduce with a decreasing beam waist or with an increasing

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degree of focusing in both the simulation and experiment. A 6-dB bandwidth in an SC-APPLN for beam waists of 50, 30 and 20 μm, according to the theory are 14 nm, 8 nm and 7 nm, compared to the experimental data which are 25 nm, 7.5 nm and 5 nm, respectively, in reasonably good agreement with the simulations. The bandwidth difference in loose focusing is due to the existence of large ripples in the theoretical SHG efficiency which decreases in the fabricated device due to random phase accumulation from imperfect poling. Further, there is a ripple-free response (for both theory and experiment) using a tightly-focused beam (ω_0 = 20μm) within the 6-dB bandwidth. Thus, the use of focused light as a novel apodization technique can increase the tolerance in fabrication for apodized SC-APPLN.

5. SH spectrum tuning by changing the focal point in the grating

When the light is focused in a specific point in the SC-APPLN with a particular period, the intensity in its Rayleigh range is increased significantly and raises the efficiency in that regime by a factor of 2 [9]. The confocal parameter is 1.62 mm for a beam waist of 20 microns which is less than the length of the single fixed period grating section. Therefore, it is expected that the efficiency will be improved within the bandwidth; however, as the intensity decreases rapidly out of the Rayleigh range for a tight focusing, the bandwidth is also reduced with respect to the bandwidth of full chirped grating.

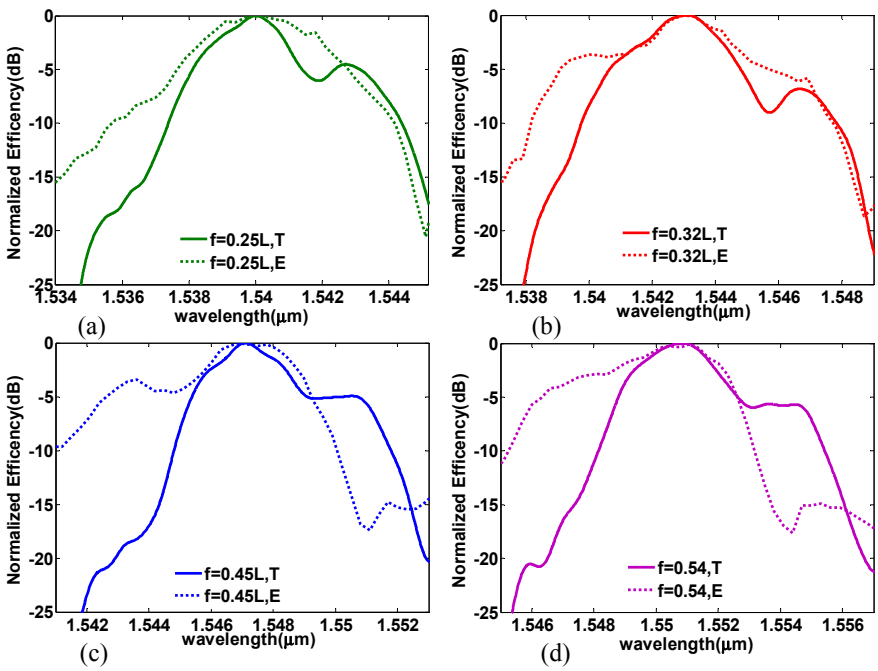

Fig. 3. Normalized measured and simulated SHG efficiency vs. FH wavelength for focused light with beam waist of 20 μm for four different focusing positions *f*, within the SC-APPLN device. In the legend "T" and "E" are theoretical and experimental data, respectively.

To convert other wavelengths available within the bandwidth of the SC-APPLN, one can simply change the focal point in the device. Thus, light is focused into a different section with a different period, resulting in the shift of the response peak. Here we swept the focal point from ¼ to almost the middle of the crystal length using a translation stage and a fixed lens with a 5-cm focal length. The normalized measured and simulated SHG efficiency versus FH wavelengths are shown in Fig. 3 (a-d) for four different focusing positions. By varying the focal point by ~5.8 mm along the length of the broadband device, the spectrum and its peak wavelength can be tuned by ~11 nm. Therefore, 1 nm tuning is possible by a 0.52 mm position displacement. The tiny changes in the shape of SHG responses in Figs. 3 (a-d) are

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due to the intrinsic non-smooth behavior of the fabricated SC-APPLN. This specially designed and fabricated device has the capacity to operate over its entire designed bandwidth. Again, the experimental results are in reasonable agreement with the experimental results. Note that the spectra remains essentially unchanged as the focal point is moved in the grating, evident from Fig. 3 (a)-(d), indicating that the apodization scheme by focusing works well even with tuning of the central SHG wavelength.

6. Conclusion

In summary, a broadband unapodized SC-APPLN was designed and carefully fabricated using a liquid-electrode poling scheme to obtain a maximum of 30-nm SHG bandwidth. The device efficiency was characterized using focused light with different beam waists. The tightly-focused Gaussian beam significantly affected the efficiency profile and suppressed the ripple in the wideband SHG response of the SC-APPLN. By increasing the degree of focusing, a ripple-free response is obtained, e.g., for a 6-dB bandwidth. For a tightly-focused beam with a beam waist of around 20 microns, the bandwidth in the unapodized SC-APPLN is >5 nm for both simulation and measurement. Furthermore, tuning the SHG spectrum over the entire SC-APPLN bandwidth is possible by displacing the focal point of the input beam inside the device. Tuning of the spectrum by 1 nm per 0.52 mm change of focal point was obtained using simulations and verified experimentally. Changing the point of focus in the crystal led to the realization of a temperature-independent tunable broadband frequency doubler, which can be used in agile multicasting of a signal via cascaded SHG and difference frequency generation (DFG) for the construction of flexible all-optical networks [2], or in the frequency doubling of high power tunable lasers, amongst other applications.

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